

all references to specific observed tokenizations hold solely for the OpenAI models.

4.7.1 Quadrant

The base grid for Quadrant was composed of “.”s separated by spaces, with an X, or other character, taking the place of a “.” where necessary, as seen in the Methodology. The new tests for quadrant can be seen in Figure 21, and the results are shown in Figures 22.

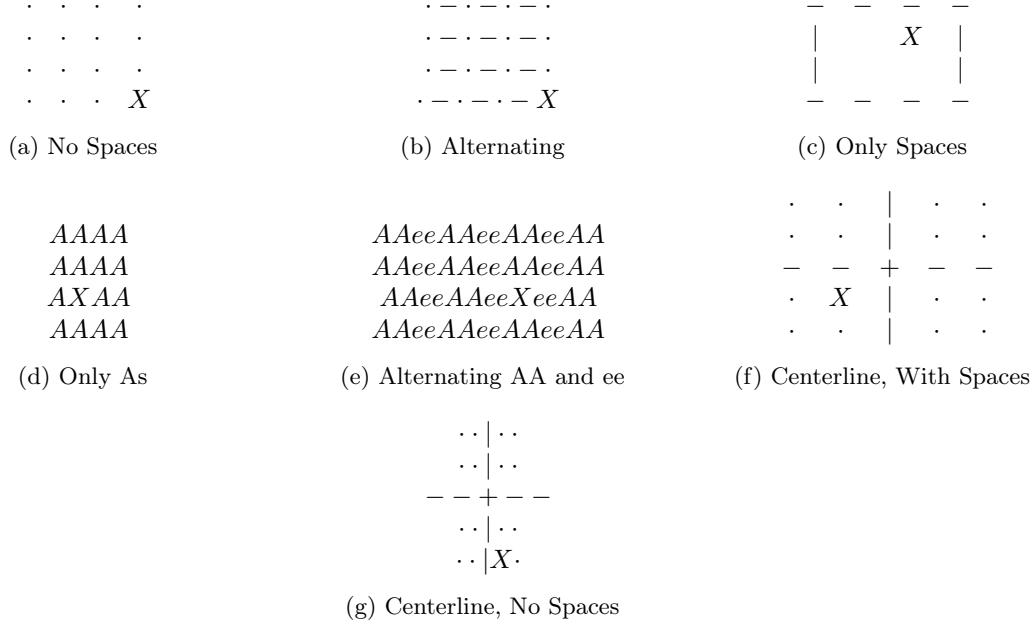


Figure 21: Types of Grids

A few of the grids—Only As, No Spaces, Only Spaces, and both Centerlines—show significant improvement over the baseline. For all tests except the centerline with spaces, a common theme emerges: the ability to tokenize large sections of a line as a singular token. In the baseline grid, each “.” was tokenized individually with the following space, which likely led to confusion and counting problems. With the no spaces grid, large sections could be put into a singular token instead, reducing the counting required and making the “X” stand out further. The upper bound of tokenization for “.”s appears to be 32 per token, but this is not regular. Smaller chunks than 32 are often split into two tokens despite being under the limit.

It perhaps seems a moot point to suggest that giving the model the center line improves performance, as it somewhat defeats the purpose of the test. Yet, there are still a few interesting observations to be gleaned from this. First, the addition of centerlines with spaces showed only moderate improvement over the baseline, implying that the models still have a difficult time identifying position in a spaced-out grid, even when given aids for the reasoning portion. Second, though, the increased accuracy implies that the models were using the centerlines as guides. This shows that they are capable of spatial reasoning in a non-strictly mathematical sense, as they could use identifiers like “left” or “above” the lines to select the quadrant. Additionally while the Anthropic models average practically 100% accuracy given the no spaces and centerline, the OpenAI models only average 80%. While this is a significant improvement, especially on larger boards, it shows that the OpenAI models still have a higher hallucination rate than the Anthropic models, as they still misplace the initial X.

It is also interesting that the test grid of alternating “AA”s and “ee”s performed worse than the baseline. The likely cause of this is that the two sets of characters always tokenize independently of one another, unlike the “.”s that tokenize with the space following. This would lead to the model needing to count and keep

track of twice as many entries, resulting in a loss in accuracy due to counting failures. The alternating test also performed worse, even though the grids were tokenized as “.-” pairs. This is probably due to a lower familiarity with such formats in the training data of the models.

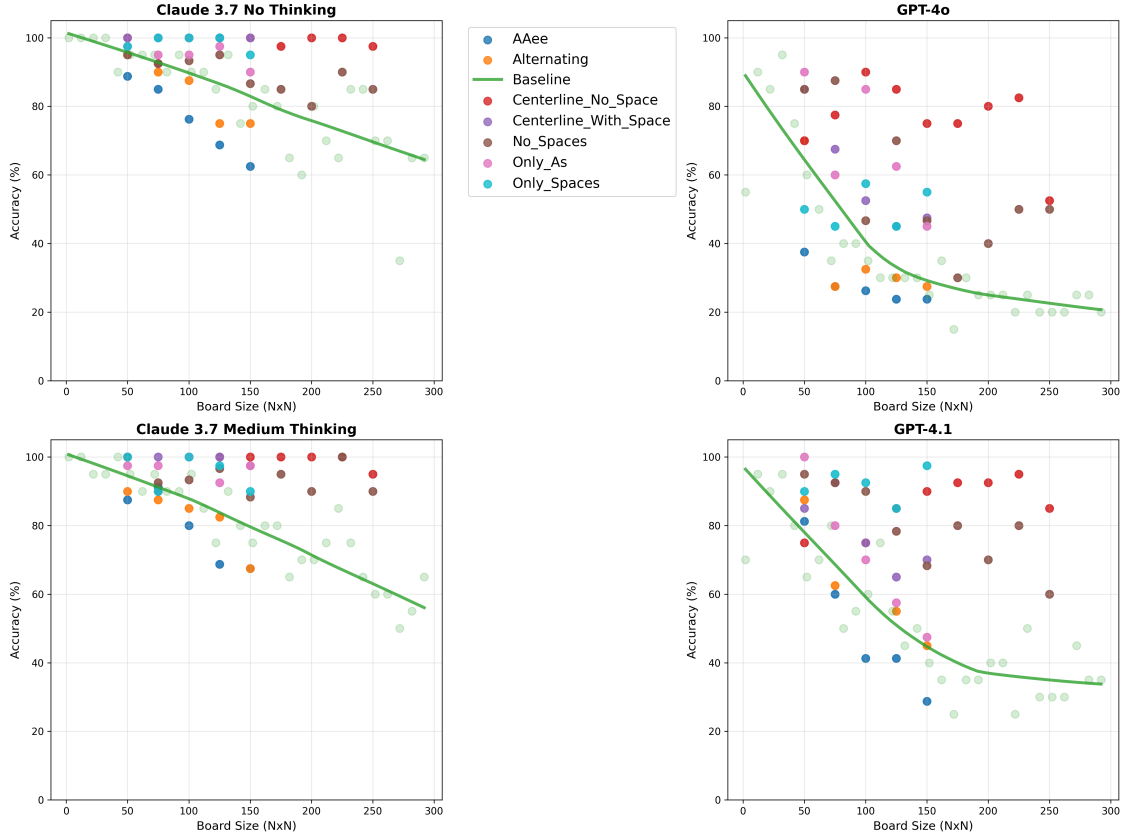


Figure 22: Accuracy vs Grid Size, New Tokenizations (Quad)

4.7.2 Search

The base grid for Search was letters separated by spaces, as seen in Methodology. The new tests for search can be seen in Figure 23, and the results can be seen in Figure 24.

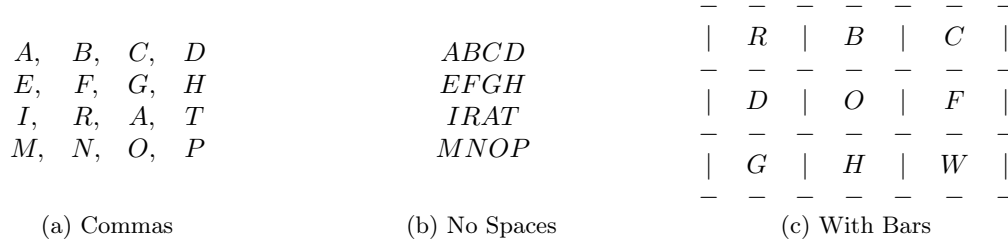


Figure 23: Types of Grids

We also trialed a non-grid based test, with the grid presented in coordinate form of $(row, column, value)$. Additionally, we tried asking the models to output more thorough step-by-step reasoning before their final answer.

As seen in the results, all tests failed to show any significant improvement over the baseline. The no spaces test, which performed significantly better in the Quadrant trials, showed a much steeper drop in accuracy here. Looking at the tokenization, in the baseline grid, each letter and the following space gets tokenized individually, like in Quadrant. However, without the spaces, the model attempts to form words from the letters, tokenizing them in small chunks. Ironically, even given a word spelled left-to-right, the tokenizer often misses it in favor of smaller groupings. It is interesting as well that step-by-step reasoning has seemingly little effect on performance. The models still skip test cases or miscount letter positions, resulting in a failed identification. These results imply that while LLMs typically excel at language processing, they are still weak in processing in non-traditional formats, such as vertically or diagonally, and are unable to reason their way through the process adequately.

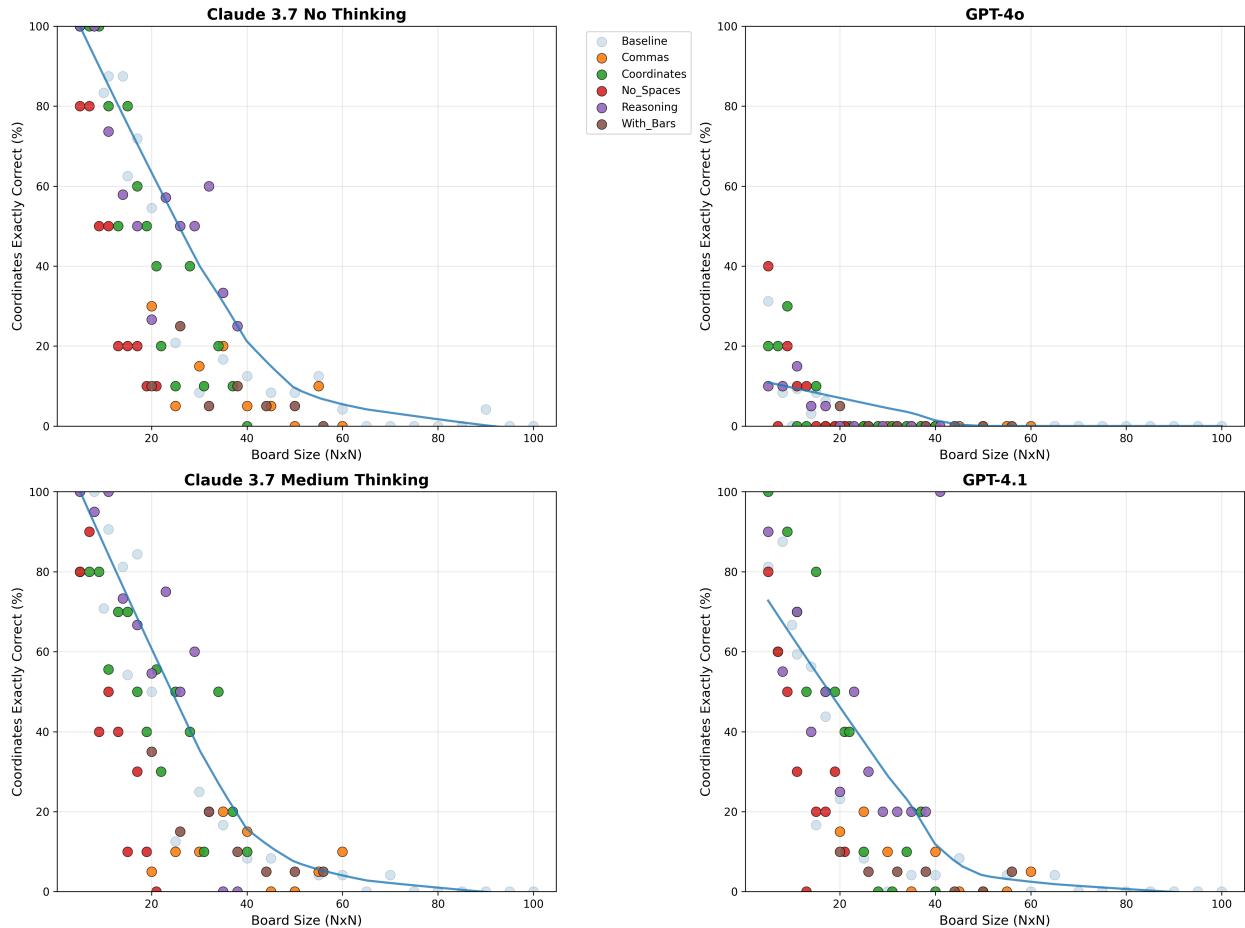


Figure 24: Accuracy vs Grid Size, New Tokenizations (Search)

5 Discussion

The results found in the above experiments underscore the fragility and unreliability of LLM spatial reasoning at scale. Across all tests and models, increased grid size led to decreased accuracy and increased error rate. Most models scored high in accuracy on smaller boards, meaning that the tests were not too complex to handle conceptually. Additionally, errors were caused by more than simple coordinate misidentification. For instance, models had difficulty calculating the center lines of the grid in the Quadrant test and made frequent